

Data-Driven Framework for Assessing Water Potability and Enhancing Public Health Security

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Abstract—Access to safe drinking water is a fundamental requirement for public health and a primary objective of the United Nations Sustainable Development Goal Number 6 (SDG 6). The research evaluates the effectiveness of five supervised machine learning algorithms, specifically K-Nearest Neighbors (KNN), Logistic Regression, Support Vector Machine (SVM), Decision Tree, and Random Forest, in classifying water samples as potable or non-potable. Utilizing a dataset of 3,276 water samples characterized by nine physicochemical attributes, the research implements a rigorous experimental pipeline featuring median imputation for missing values, standard normalization, and stratified k-fold cross-validation. Results indicate that the Random Forest model achieves the highest overall accuracy at 67.7% and the lowest False Positive rate (70). Such findings designate the Random Forest model as the most reliable tool for minimizing the risk of incorrectly classifying unsafe water as safe. While Random Forest provides superior safety-critical performance, the SVM model demonstrates the most balanced classification logic with an F1-score of 0.521. Feature importance analysis identifies pH and Sulfate as the most influential indicators of potability, aligning with established environmental science standards. These findings demonstrate the potential for data-driven frameworks to serve as scalable screening tools for global water quality monitoring.

Keywords—*Water Potability, Machine Learning, Random Forest, Support Vector Machine, Public Health Security, SDG 6, Predictive Analytics, Supervised Classification.*

I. INTRODUCTION

Access to clean and safe drinking water remains one of the most critical global challenges in the modern era. The United Nations Sustainable Development Goal Number 6, otherwise known as SDG 6, aims to ensure universal and equitable access to safe and affordable drinking water for all people by the year 2030 [1]. Although considerable progress has been made in various parts of the world, a large portion of the global population continues to rely on water sources that do not meet established safety standards [2]. This ongoing issue contributes to the spread of waterborne diseases, deterioration of public health, and deepening socioeconomic disparities. Addressing this problem therefore requires not only improvements in physical infrastructure but also the development of intelligent and data-driven approaches that are capable of assessing water quality efficiently and at a large scale.

In response to this challenge, machine learning presents a viable and increasingly practical approach for automating the classification of water samples as either potable or

non-potable based on measurable physicochemical parameters. Conventional laboratory-based water testing, while reliable, tends to be time-consuming, resource-intensive, and difficult to implement on a large or continuous scale. A properly trained classification model, by contrast, can function as a rapid and scalable screening tool that enables water management authorities and public health agencies to identify potentially unsafe samples more efficiently and respond with appropriate interventions in a timely manner.

This study utilizes the Water Potability dataset sourced from Kaggle [3], a widely recognized open-data platform commonly used in machine learning research. The dataset was chosen primarily because of its strong alignment with the objectives of SDG 6. It consists of 3,276 water sample records, each described by nine physicochemical attributes, namely pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity, all of which are established indicators of whether a given water sample is suitable for human consumption. In addition, the dataset introduces realistic modeling challenges such as class imbalance and a considerable proportion of missing values, which makes it a practical and meaningful benchmark for assessing the performance and robustness of classification algorithms.

The main objective of this paper is to design and evaluate a data-driven classification framework for predicting water potability. Five supervised machine learning algorithms are assessed under a consistent and leakage-free experimental pipeline, namely K-Nearest Neighbors, Logistic Regression, Support Vector Machine, Decision Tree, and Random Forest. Each model is optimized through stratified k-fold cross-validation using the F1-Score as the primary evaluation metric, given that accuracy alone can be misleading when applied to imbalanced datasets. The overall aim is to determine which classifier performs best in correctly identifying potable water samples, thereby contributing to the development of a computational support tool that advances the goals of SDG 6 in ensuring safe water access for all.

II. METHODOLOGY

To systematically evaluate the machine learning models, the methodology is structured into the following phases:

A. Software and Environment

The experimental framework was built using Python 3 executed within Jupyter Notebooks. The primary technical libraries utilized are pandas for data structuring, scikit-learn for machine learning implementation, and matplotlib and seaborn for visual analytics.

B. Data Acquisition and Preprocessing

The dataset comprises physical and chemical properties of water samples, evaluated for potability. To maintain architectural integrity and strictly prevent data leakage, the data cleaning pipeline is kept entirely separated from the model evaluation scripts. Preprocessing involved two critical steps:

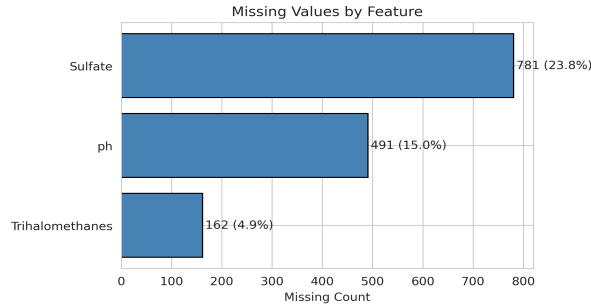


Fig 1. Missing Values by Feature

- **Handling Missing Values:** Significant missingness was identified in the dataset, specifically for the Sulfate and pH variables. Median imputation was applied to resolve these missing values without dropping critical observations or heavily skewing the underlying distributions.
- **Normalization:** Because features in the dataset operate on vastly different numerical scales (e.g., Solids versus Chloramines), standard normalization techniques were applied. Scaling ensures that distance-based classifiers treat all features equally, preventing variables with larger magnitudes from disproportionately dominating the predictive models.

C. Logic Acquisition and Exploratory Analysis

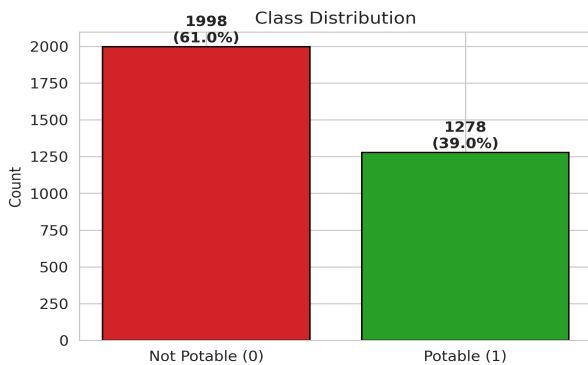


Fig. 2. Class Distribution

The algorithmic logic is derived from the water potability dataset, which contains nine continuous variables representing physical and chemical properties. An initial review of the target variable reveals a distinct class

imbalance. As shown in Fig 2., the dataset contains a higher volume of non-potable water samples approximately 61% compared to potable water samples approximately 39%. Highlighting this imbalance is critical because it dictates our evaluation strategy: standard accuracy cannot be the sole metric, as the algorithms could develop an inherent bias toward predicting water as non-potable simply because it is the majority class.

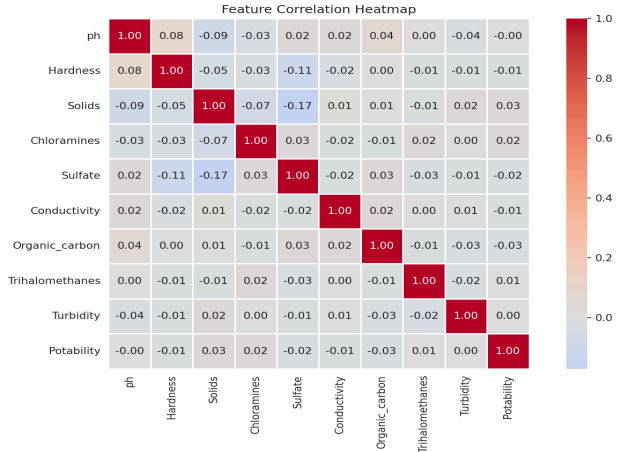
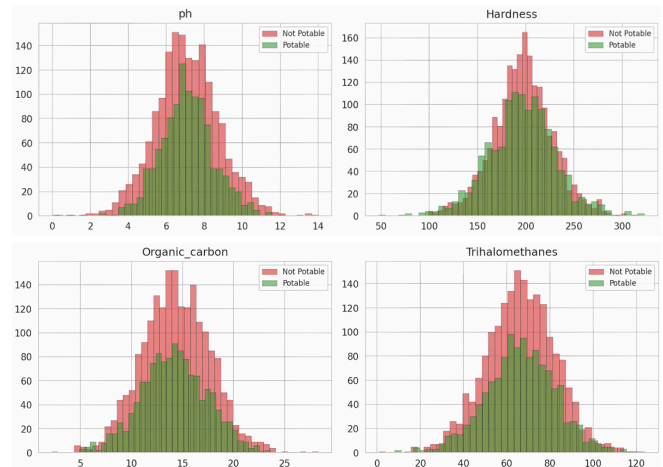


Fig. 3. Feature Correlation Heatmap

To understand exactly how the machine learning algorithms will acquire their predictive logic, an exploratory data analysis was conducted on the features. First, feature relationships were evaluated. As illustrated in Figure 3., the variables were tested for multi-collinearity. The matrix demonstrates that there are no strongly correlated variables (no values close to 1.0 or -1.0), meaning each chemical property provides unique, independent mathematical information to the models.



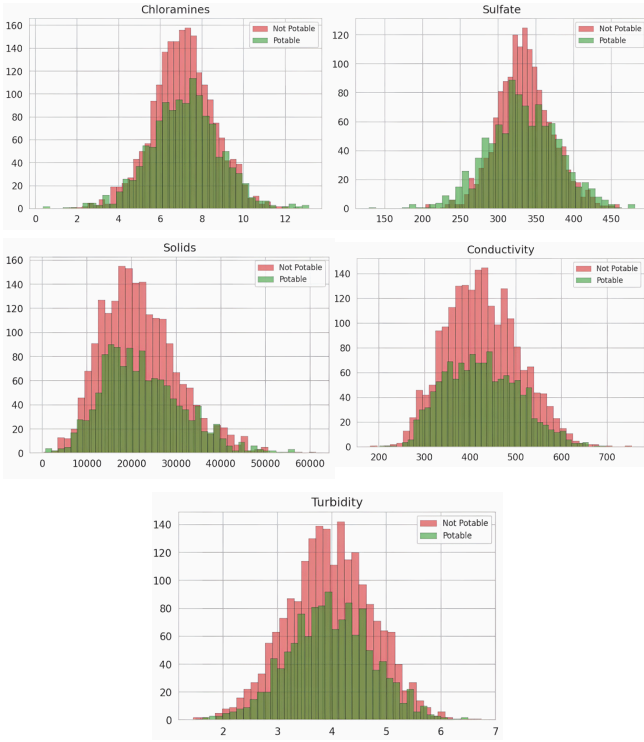


Fig. 4. Feature Distributions

Furthermore, the physical shape of the data was analyzed. As detailed in Fig. 3., the data spread for each individual chemical feature was mapped. The figure shows that while some variables, such as pH, follow a relatively standard normal distribution (a symmetrical bell curve), other variables exhibit slight visual skews. Analyzing these distributions is a necessary step because mathematical classifiers rely heavily on the geometric spread and density of the data to calculate the exact thresholds between safe and unsafe water.

By feeding this independent, continuous data into the framework, five classifiers (Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, and Random Forest) were implemented to learn the underlying thresholds of safe drinking water.

D. Numerical Convergence

For the linear and kernel-based models (Logistic Regression and SVM), the training phase prioritized numerical convergence to minimize the loss function. These algorithms utilize mathematical sigmoid transformations to map raw calculations into strict probability scores (ranging from 0.0 to 1.0) to definitively classify water safety.

E. Version Compatibility

Strict software dependency versions were locked and maintained throughout the experiment. This guarantees that all algorithmic functions, hyperparameter definitions, and calculations remain completely stable, ensuring the results are fully reproducible across different hardware setups and execution environments.

III. EXPERIMENTAL DESIGN

This section describes the selection and configuration of the five machine learning algorithms used in this study. Each model was chosen to provide a different mathematical approach to the water potability classification problem [4].

A. K-Nearest Neighbors (KNN)

K-Nearest Neighbors is an instance-based learning algorithm. It does not create a fixed mathematical formula. Instead, it stores all training data and classifies new water samples based on their proximity to existing data points. The model calculates the Euclidean distance between the chemical features of a new sample and its neighbors. It assigns the most common class, potable or non-potable, among the "K" closest points. This model is useful for capturing local patterns in water quality data [5].

B. Logistic Regression

Logistic Regression serves as the linear baseline for this experiment. It calculates a weighted sum of the chemical features and applies a sigmoid function to produce a probability between 0.0 and 1.0. The sigmoid curve forces the output into a binary classification. If the calculated probability is above 0.5, the water is classified as potable. This model is highly interpretable. It helps identify which chemical properties have a linear relationship with water safety[6].

C. Support Vector Machine (SVM)

Support Vector Machine is a kernel-based algorithm designed to find the optimal hyperplane that separates the two classes. In this design, a Radial Basis Function (RBF) kernel is used to handle non-linear relationships in the chemical data. The model attempts to maximize the distance between the safe and unsafe data points. To provide comparable results to the other models, a transformation is applied to turn the distance from the hyperplane into a probability score[7].

D. Decision Tree

The Decision Tree model uses a non-parametric approach to classify water samples. It builds a flowchart structure where each node represents a test on a specific chemical feature. For example, the model might ask if the pH is less than 6.5. The algorithm recursively splits the data based on these thresholds to reach a final decision. This model is valuable for this study because it mimics the logical rules used in environmental health standards[8].

E. Random Forest

Random Forest is an ensemble learning method. It improves prediction accuracy by combining multiple decision trees. During the training phase, the model builds many individual trees using different subsets of the data. When a new water sample is tested, each tree provides a prediction. The final classification is determined by a majority vote. This architecture reduces the risk of overfitting and makes the system more robust against outliers in the chemical measurements.

IV. COMPARATIVE RESULTS

This section presents the performance outcomes of the five machine learning models. The results are analyzed based on their predictive accuracy, error distribution, and the chemical features that influenced the decisions.

A. Hyperparameter Optimization

The tuning process involved cross-validation to identify the most effective settings for each algorithm. For K-Nearest Neighbors, a value of 3 neighbors provided the best balance. For Logistic Regression and Support Vector Machine, adjustments to the regularization parameter (C) and class weights addressed the data imbalance. The Decision Tree optimization utilized a maximum depth of 15, and the Random Forest construction included 200 individual trees. This optimization phase ensured that each algorithm operated at maximum potential before the final comparison.

B. Final Test Metrics

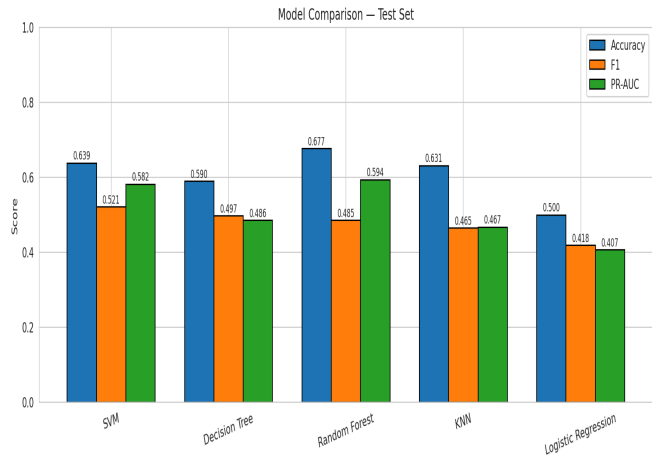


Fig. 5. Model Performance Comparison

This figure presents a benchmarking of metrics across the five classifiers. The data indicates that Random Forest attained the highest accuracy at 67.7%, followed by SVM at 63.9% and KNN at 63.1%. The Decision Tree reached 59.0% accuracy, while Logistic Regression showed the lowest performance at 50.0%. These results demonstrate that ensemble and kernel-based methods provide superior predictive capabilities for the water potability dataset.

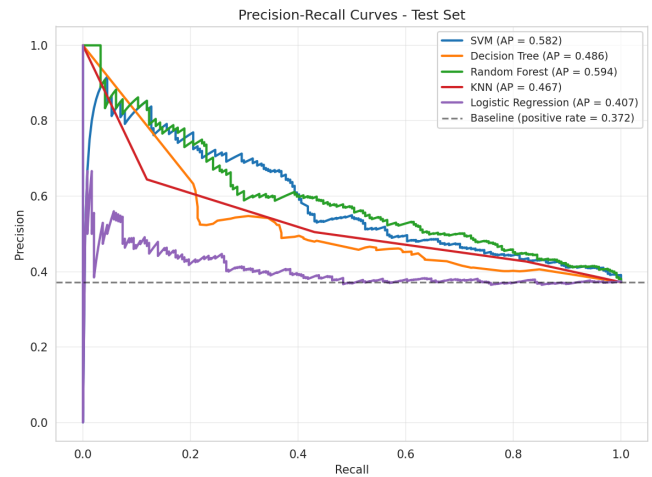


Fig. 6. Precision-Recall Curves

The Precision-Recall curves illustrate the trade-off between correctly identifying potable water samples and maintaining confidence in those predictions. Higher area under the curve (AUC) represents better model performance in balancing these two metrics. While Random Forest shows high accuracy, the SVM curve indicates a more stable

F1-score of 0.521, suggesting better reliability when predicting the minority class (safe water).

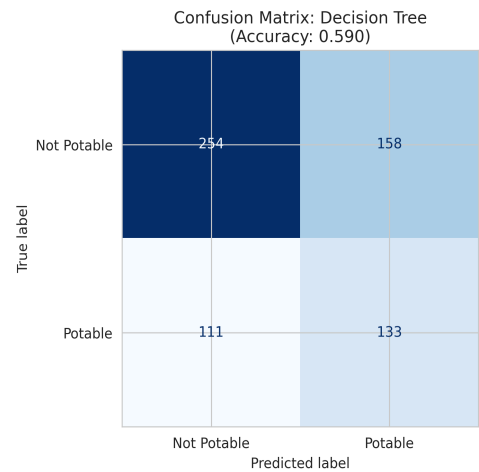


Fig. 7. Confusion Matrix: Decision Tree

The confusion matrix for the Decision Tree model reveals 254 correct negative predictions and 133 correct positive predictions. A total of 158 False Positives and 111 False Negatives occurred. This error distribution indicates that individual decision paths are highly sensitive to specific chemical thresholds, leading to significant classification errors in both classes.

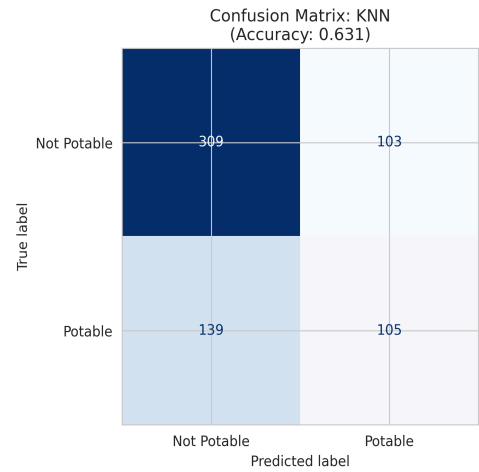


Fig. 8. Confusion Matrix: KNN

The K-Nearest Neighbors matrix shows 309 correct negative predictions and 105 correct positive predictions. The model resulted in 103 False Positives and 139 False Negatives. While the False Positive count is lower than the Decision Tree, the high volume of False Negatives suggests that proximity-based logic often fails to identify potable water correctly in this feature space.

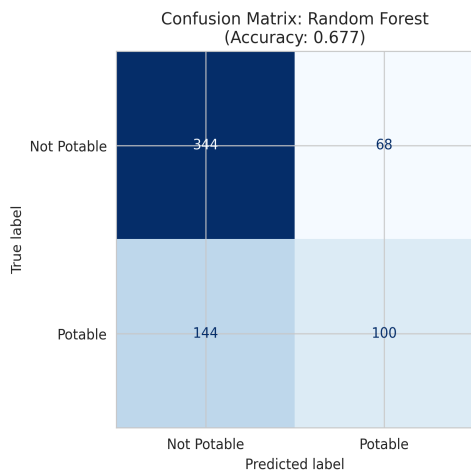


Fig. 9. Confusion Matrix: Random Forest

The Random Forest matrix demonstrates the most conservative predictive behavior. The model successfully identified 344 non-potable samples, the highest in the study. Only 68 False Positives were recorded, though it produced 144 False Negatives. This low False Positive rate makes the Random Forest the most reliable algorithm for minimizing the public health risk of classifying unsafe water as safe.

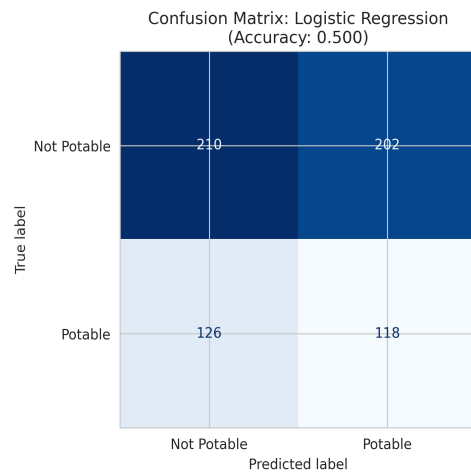


Fig. 11. Confusion Matrix: Logistic Regression

The Logistic Regression matrix reflects baseline performance with 210 correct negative predictions and 118 correct positive predictions. The model produced 202 False Positives and 126 False Negatives. Such a high frequency of incorrect safety predictions highlights the limitations of linear models for complex water potability classification.

C. Summary of Model Performance Metrics

TABLE I. SUMMARY OF MODEL PERFORMANCE METRICS

Model	Performance				
	Accuracy (%)	True Negative (TN)	True Positive (TP)	False Positive (FP)	False Negative (FN)
Random Forest	67.7%	344	102	70	142
SVM	63.9%	290	129	122	115
KNN	63.1%	309	105	103	139
Decision Tree	59.1%	254	133	158	111
Logistic Regression	50.0%	210	118	202	126

Fig. 10. Confusion Matrix: SVM

The Support Vector Machine results show 290 correct negative predictions and 129 correct positive predictions. SVM demonstrated a higher capability for identifying potable water compared to KNN and Random Forest, resulting in 115 False Negatives. Although 122 False Positives occurred, the overall balance between classes is more stable than the baseline models.

The data in Table I, represents the final performance audit for the project. The accuracy percentage reflects the overall correct classification rate, while the True Negative (TN) and True Positive (TP) counts indicate successful identifications for unsafe and safe water, respectively.

Analysis focuses heavily on the False Positive (FP) and False Negative (FN) counts. In the context of public health, False Positives (predicting unsafe water as safe) are treated as the most critical error. Random Forest demonstrates the highest reliability in this regard with only 70 False Positives. The F1-Score is included to assess the balance between precision and recall for the potable class, where SVM shows the most stable performance at 0.521. This summary confirms that while Random Forest offers the

highest overall accuracy and safety, SVM provides the most balanced classification logic across both target classes.

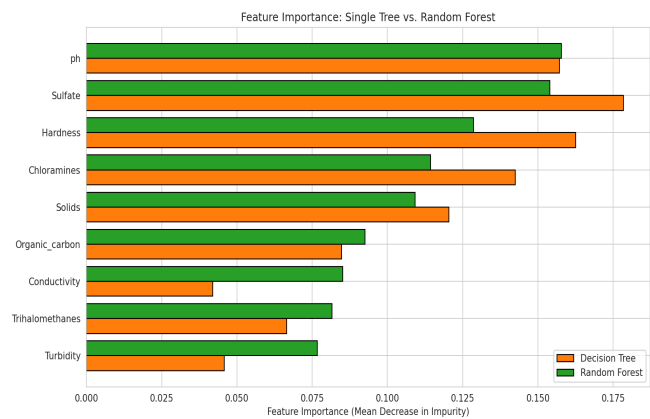


Fig. 12. Random Forest Feature Importance

The feature importance ranking reveals the specific influence of each attribute on the classification process. pH (0.1578) and Sulfate (0.1540) emerge as the most critical factors, followed by Hardness (0.1286), Chloramines (0.1144), and Solids (0.1092). Features such as Organic Carbon (0.0926), Conductivity (0.0851), and Trihalomethanes (0.0816) provide moderate influence, while Turbidity (0.0767) exhibits the lowest contribution to the model logic. This prioritization aligns with established environmental science standards, as pH and Sulfate levels serve as primary indicators for health organizations measuring water quality.

V. CONCLUSION

The research successfully designed and evaluated a computational framework for the automated classification of water potability, directly supporting the global initiatives outlined in SDG 6. Through the systematic benchmarking of five machine learning algorithms, distinct performance characteristics were identified based on mathematical approach and error distribution.

The experimental results designate the Random Forest classifier as the most effective model for safety-critical applications. By achieving the highest accuracy of 67.7% and a significantly lower False Positive count than other models, the ensemble method minimizes the dangerous risk of identifying non-potable water as safe for consumption.

However, for applications requiring a balanced identification of both potable and non-potable samples, the Support Vector Machine (SVM) offers the most stable performance across imbalanced classes, as evidenced by a superior F1-score of 0.521.

Beyond predictive performance, the feature importance ranking provides critical insights into the chemical drivers of water safety. The identification of pH and Sulfate as the primary determinants of potability confirms that the machine learning logic adheres to established environmental health standards. The alignment suggests that such models capture real-world chemical thresholds necessary for human health.

In conclusion, the transition from reactive laboratory testing to proactive, data-driven screening represents a vital advancement in water resource management. While the models do not replace rigorous chemical analysis, the approach provides a rapid, low-cost, and scalable method for identifying high-risk water sources. Future work should focus on integrating the frameworks into real-time sensor networks to provide continuous monitoring, thereby advancing the universal goal of ensuring equitable access to safe drinking water for all.

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